

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE****Regular Winter Examination – 2024****Course: B.Tech Branch : Electronics & Computer Engg/Electronics & Computer Science****Semester :VII****Subject Code & Name: BTECOE705B Cyber Security & Block chain Technology****Max Marks: 60****Date: 21/02/2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

|             |                                                                            | (Level/CO)                         | Marks                                                 |                                                                               |
|-------------|----------------------------------------------------------------------------|------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Q. 1</b> | <b>Objective type questions. (Compulsory Question)</b>                     |                                    | <b>12</b>                                             |                                                                               |
| 1           | Cryptanalysis is used to _____.                                            | 2                                  | 1                                                     |                                                                               |
|             | a. find some insecurity in a cryptographic scheme                          | b. to increase the speed           | c. to encrypt the data                                | d. to make new ciphers                                                        |
| 2           | Which of the following algorithm does not belong to symmetric key ciphers? | 2                                  | 1                                                     |                                                                               |
|             | a.Rc4                                                                      | b. DES                             | c. Diffie-Hellman                                     | d. AES                                                                        |
| 3           | Which of the following is incorrect about symmetric key cryptography?      | 1                                  | 1                                                     |                                                                               |
|             | a. It uses common key for both encryption and decryption                   | b. It is a secret key cryptography | c. It uses single fixed length key for each algorithm | d. Symmetric key can be combined with asymmetric key to achieve more security |
| 4           | Electronic Code Book process is used in _____.                             | 1                                  | 1                                                     |                                                                               |
|             | a. caesar cipher                                                           | b. antisymmetric cipher            | c. block cipher                                       | d. stream cipher                                                              |
| 5           | _____ is an example of asymmetric ciphers.                                 | 3                                  | 1                                                     |                                                                               |

|     |                                                                                                           |                                                               |                               |                               |   |    |
|-----|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------|-------------------------------|---|----|
|     | a. Block cipher                                                                                           | b. RSA                                                        | c. AES                        | d. Advanced cryptology        |   |    |
| 6   | What type of algorithm does AES encryption use?                                                           |                                                               |                               |                               | 3 | 1  |
|     | a. Stream cipher                                                                                          | b. Symmetric block cipher                                     | c. Asymmetric caesar cipher   | d. DES encryption             |   |    |
|     | What is the first block of a Blockchain called?                                                           |                                                               |                               |                               | 5 | 1  |
|     | a. Seed Block                                                                                             | b. Origin Block                                               | c. Genesis Block              | d. Root Block                 |   |    |
|     | What is the unit structure of a Blockchain called?                                                        |                                                               |                               |                               | 5 | 1  |
|     | a. Node                                                                                                   | b. Transaction                                                | c. Block                      | d. Record                     |   |    |
| 9   | When a hash function is used to provide message authentication, the hash function value is referred to as |                                                               |                               |                               | 4 | 1  |
|     | a. Message Field                                                                                          | b. Message Digest                                             | c. Message Score              | d. Message Leap               |   |    |
| 10  | Message authentication code is also known as                                                              |                                                               |                               |                               | 4 | 1  |
|     | a. key code                                                                                               | b. hash code                                                  | c. keyed hash function        | d. message key hash function  |   |    |
| 11  | Which of the following actions compromise cyber security?                                                 |                                                               |                               |                               | 1 | 1  |
|     | a. Vulnerability                                                                                          | b. Attack                                                     | c. Threat                     | d. Exploit                    |   |    |
| 12  | What is the primary purpose of Blockchain technology?                                                     |                                                               |                               |                               | 5 | 1  |
|     | a. To replace all traditional databases                                                                   | b. To provide a secure and transparent record of transactions | c. To create cryptocurrencies | d. To centralize data storage |   |    |
| Q.2 | Solve the following.                                                                                      |                                                               |                               |                               |   | 12 |
| A)  | Compare stream cipher with block cipher with example.                                                     |                                                               |                               |                               | 2 | 6  |
| B)  | Elaborate working of Blockchain technology.                                                               |                                                               |                               |                               | 5 | 6  |
| Q.3 | Solve the following.                                                                                      |                                                               |                               |                               |   | 12 |

|             |                                                                                  |   |           |
|-------------|----------------------------------------------------------------------------------|---|-----------|
| A)          | With the help of block diagram AES algorithm in detail.                          | 3 | 6         |
| B)          | What is data integrity, non-repudiation and authentication?                      | 1 | 6         |
|             |                                                                                  |   |           |
| <b>Q. 4</b> | <b>Solve Any Two of the following.</b>                                           |   | <b>12</b> |
| A)          | Describe Elliptic Curve Cryptography algorithm.                                  | 4 | 6         |
| B)          | Elaborate RSA algorithm with example.                                            | 2 | 6         |
|             | What is HMAC?                                                                    | 2 | 6         |
|             |                                                                                  |   |           |
| <b>Q.5</b>  | <b>Solve Any Two of the following.</b>                                           |   | <b>12</b> |
| A)          | Describe SSL in detail.                                                          | 3 | 6         |
| B)          | Explain the encryption and decryption of DES algorithm in detail.                | 3 | 6         |
| C)          | Compare public Blockchain with private Blockchain.                               | 5 | 6         |
|             |                                                                                  |   |           |
| <b>Q. 6</b> | <b>Solve Any Two of the following.</b>                                           |   | <b>12</b> |
| A)          | Identify the possible threats for RSA algorithm and list their counter measures. | 3 | 6         |
| B)          | Write a note on classical cryptosystems.                                         | 1 | 6         |
| C)          | How firewalls provides security?                                                 | 4 | 6         |
|             | <b>*** End ***</b>                                                               |   |           |